The loss of codiversified gut microbes predicts Autism Spectrum Disorder status and severity

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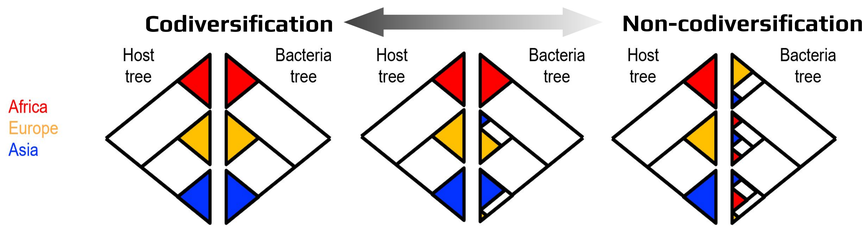
Background

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition with rising diagnosis rates in the United States. Diagnosis is based on behavioral criteria, as there is currently no biochemical test. Severe cases of ASD are often accompanied by gastrointestinal symptoms, suggesting involvement of the gut–brain axis — the bidirectional communication between the gut and the brain.

Humans and some of their gut microbial species exhibit phylogenetic congruence with their hosts (co-diversification), showing human ancestry-specific strains that are maintained through transmission among genetically related individuals for thousands of years (Suzuki et al. 2022). Although several known co-diversifying microbes have been reported to be depleted in individuals with ASD, whether the loss of co-diversified species is linked to ASD disease status and severity remains unexplored.



Question

Does the loss of co-diversified microbes predict ASD status and severity?

Methods

- Fecal shotgun metagenomic data was generated before and after a double-blind, placebo-controlled microbiota transplant therapy (MTT) trial.
- Published list of globally prevalent 36 codiversified and 23 non-codiversified microbial species was used from Suzuki et al. 2022.
- Childhood Autism Rating Scale (CARS) scores was used to measure ASD severity

Phase 2 Microbiota Transplant Therapy (MTT) trial

ASD subjects (n= 51)

Before MTT (Time point 1) | **After MTT** (Time point 2)

Typically developing (TD) (n =46)

No intervention (Time point 1)

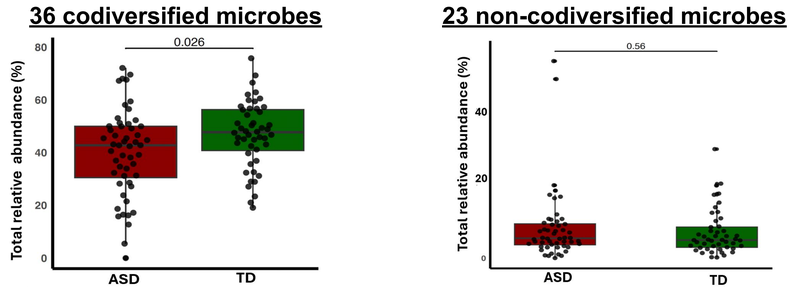
The study was approved by ASU IRB# STUDY00020527

Reference

Suzuki TA, et al. Codiversification of gut microbiota with humans. Science. 2022 Sep 16; 377(6612):1328-1332.

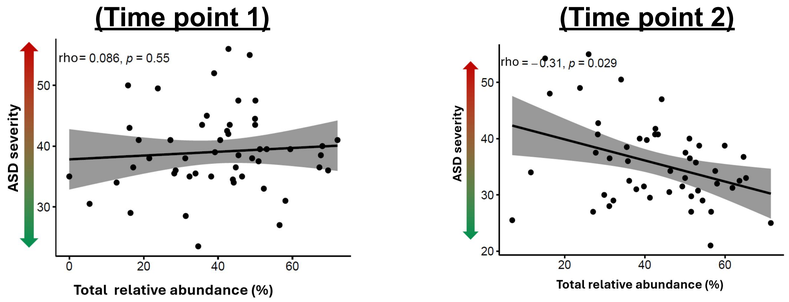
Results

Adults with ASD have a lower total relative abundance of codiversified microbes compared to typically developing (TD) controls. (Wilcoxon rank sum test)



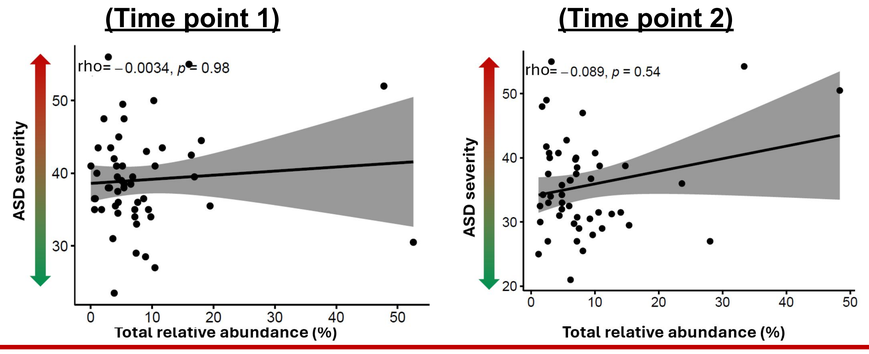
A decrease in ASD severity after MTT is associated with an increase in the total relative abundance of codiversified microbes. (Spearman rho)

36 codiversified microbes



The total relative abundance of non-codiversified microbes do not predict ASD severity (Spearman rho)

23 Non-codiversified microbes



Conclusion

Codiversifying microbes can serve as biomarkers of ASD status and severity

Future directions

Use combinations of self-identified participant ancestry and microbial ancestry to predict ASD status, severity, and MTT outcomes.

Acknowledgements

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